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Physics 180 THV-1
Problem Set 8-9

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Problem 8.1

Particle A , at rest decays into particles B and C :

$$A \rightarrow B + C \tag{1.1}$$

What are the energies E_B and E_C of the outgoing particles B and C in terms of the various masses? What are the magnitudes of the outgoing momenta $|p_B|$ and $|p_C|$ (also in terms of the masses)? As a test to see if your answer is correct, $|p_B|$ should go to zero when $m_A = m_B + m_C$, and becomes imaginary when $m_A < (m_B + m_C)$. Explain why these behaviors happen.

Solution

In the rest frame of A , four-momentum conservation gives

$$p_A^\mu = p_B^\mu + p_C^\mu$$

with

$$p_A^\mu = (m_A, \mathbf{0})$$

in natural units $c = 1$. Since the outgoing momenta must be equal and opposite,

$$p_B = -p_C, \quad |p_B| = |p_C| \equiv p$$

The energies are therefore

$$E_B = \sqrt{p^2 + m_B^2}, \quad E_C = \sqrt{p^2 + m_C^2}$$

and energy conservation requires

$$E_B + E_C = m_A$$

Solving for this gives

$$E_B = \frac{m_A^2 + m_B^2 - m_C^2}{2m_A}$$

and

$$E_C = \frac{m_A^2 + m_C^2 - m_B^2}{2m_A}$$

The common magnitude of the outgoing momentum is

$$p = \sqrt{E_B^2 - m_B^2}$$

Substituting the expression for E_B , we have

$$p = \frac{1}{2m_A} \sqrt{(m_A^2 - (m_B + m_C)^2)(m_A^2 - (m_B - m_C)^2)}$$

Thus

$$|p_B| = |p_C| = \frac{\sqrt{(m_A^2 - (m_B + m_C)^2)(m_A^2 - (m_B - m_C)^2)}}{2M_A}$$

The threshold behaviour, i.e., if $m_A = m_B + m_C$, then

$$m_A^2 - (m_B^2 + m_C^2)^2 = 0$$

so

$$p = 0$$

Physically, this is the threshold for decay. The entire rest energy of A is just enough to provide the rest masses of B and C . There is no energy left for kinetic energy.

If $m_A < m_B + m_C$, then

$$m_A^2 - (m_B^2 + m_C^2)^2 < 0$$

while the other factor remains positive for positive masses in the relevant region. Consequently

$$p^2 < 0$$

and p becomes imaginariy. This means there is no physical kinematic solution. A does not have enough invariant mass to produce $B + C$, therefore the decay is forbidden.

Problem 9.1

Construct another explicit representation of the α and β matrices different from the Dirac-Pauli representation and show that they satisfy the anti-commutation relations

$$\alpha_x^2 = \alpha_y^2 = \alpha_z^2 = \beta^2 = \mathbb{1} \quad (2.1)$$

$$\alpha_j \beta + \beta \alpha_j = 0 \quad (2.2)$$

$$\alpha_j \alpha_k + \alpha_k \alpha_j = 0 \quad (j \neq k) \quad (2.3)$$

How are your new representations related to the Dirac-Pauli representation? Does the specific form of the Dirac-Pauli representation offer any advantages (if any) to the one you've constructed and versus any other representations in general?

Solution

The relations require the matrices to have even dimension. Since

$$\beta^2 = \mathbb{1},$$

the eigenvalues of β can only be $+1$ and -1 . Let V_+ and V_- denote the corresponding eigenspaces. If $\psi_+ \in V_+$, then

$$\beta \psi_+ = \psi_+.$$

Using

$$\alpha_j \beta + \beta \alpha_j = 0,$$

we obtain

$$\beta(\alpha_j \psi_+) = -\alpha_j \beta \psi_+ = -\alpha_j \psi_+.$$

Thus, α_j maps V_+ into V_- . Similarly, it maps V_- into V_+ . Since

$$\alpha_j^2 = \mathbb{1},$$

each α_j is invertible, so these mappings must be one-to-one in both directions. Therefore,

$$\dim V_+ = \dim V_-.$$

If each eigenspace has dimension n , then the total dimension is

$$N = \dim V_+ + \dim V_- = 2n.$$

Hence the matrices can only have even dimension:

$$N = 2, 4, 6, \dots$$

In particular, a 3×3 representation is impossible.

Although 2×2 is allowed by this argument, it is not large enough to represent all four matrices $\alpha_x, \alpha_y, \alpha_z, \beta$. The three Pauli matrices already provide three mutually anticommuting 2×2 matrices,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

which satisfy

$$\sigma_j^2 = \mathbb{1}, \quad \sigma_j \sigma_k + \sigma_k \sigma_j = 0 \quad (j \neq k).$$

However, there is no fourth nonzero 2×2 matrix that anticommutes with all three Pauli matrices while also satisfying $M^2 = \mathbb{1}$. Thus 2×2 is insufficient for the full set of four matrices. The next possible dimension is therefore 4×4 , which is the smallest nontrivial representation of the Dirac algebra.

One representation different from the Dirac–Pauli representation is the Weyl, or chiral, representation[2]. Define

$$\alpha_j = \begin{pmatrix} \sigma_j & 0 \\ 0 & -\sigma_j \end{pmatrix}, \quad \beta = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}.$$

Explicitly,

$$\begin{aligned} \alpha_x &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, & \alpha_y &= \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & -i & 0 \end{pmatrix}, \\ \alpha_z &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, & \beta &= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Since the Pauli matrices satisfy $\sigma_j^2 = \mathbb{1}$,

$$\alpha_j^2 = \begin{pmatrix} \sigma_j^2 & 0 \\ 0 & \sigma_j^2 \end{pmatrix} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix} = \mathbb{1}_4.$$

Furthermore,

$$\beta^2 = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix} = \mathbb{1}_4.$$

Therefore,

$$\alpha_x^2 = \alpha_y^2 = \alpha_z^2 = \beta^2 = \mathbb{1}_4.$$

For the anticommutation relation between α_j and β ,

$$\alpha_j \beta = \begin{pmatrix} \sigma_j & 0 \\ 0 & -\sigma_j \end{pmatrix} \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{pmatrix},$$

while

$$\beta \alpha_j = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} \begin{pmatrix} \sigma_j & 0 \\ 0 & -\sigma_j \end{pmatrix} = \begin{pmatrix} 0 & -\sigma_j \\ \sigma_j & 0 \end{pmatrix}.$$

Hence,

$$\alpha_j \beta + \beta \alpha_j = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0.$$

For $j \neq k$,

$$\alpha_j \alpha_k = \begin{pmatrix} \sigma_j & 0 \\ 0 & -\sigma_j \end{pmatrix} \begin{pmatrix} \sigma_k & 0 \\ 0 & -\sigma_k \end{pmatrix} = \begin{pmatrix} \sigma_j \sigma_k & 0 \\ 0 & \sigma_j \sigma_k \end{pmatrix},$$

and

$$\alpha_k \alpha_j = \begin{pmatrix} \sigma_k \sigma_j & 0 \\ 0 & \sigma_k \sigma_j \end{pmatrix}.$$

Since

$$\sigma_j \sigma_k + \sigma_k \sigma_j = 0 \quad (j \neq k),$$

we have

$$\alpha_j \alpha_k + \alpha_k \alpha_j = \begin{pmatrix} \sigma_j \sigma_k + \sigma_k \sigma_j & 0 \\ 0 & \sigma_j \sigma_k + \sigma_k \sigma_j \end{pmatrix} = 0.$$

Thus the Weyl representation satisfies all of the required relations.

The Dirac–Pauli representation is

$$\alpha_j^{(D)} = \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix}, \quad \beta^{(D)} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}.$$

The Weyl and Dirac–Pauli representations are related by a unitary change of basis. In particular, with

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbb{1} & \mathbb{1} \\ -\mathbb{1} & \mathbb{1} \end{pmatrix},$$

the Weyl matrices are obtained from the Dirac–Pauli matrices through

$$\alpha_j^{(W)} = U \alpha_j^{(D)} U^\dagger, \quad \beta^{(W)} = U \beta^{(D)} U^\dagger.$$

Thus, the two representations contain the same physical information and differ only in the choice of basis used to represent the Dirac spinors.

The particular form of the Dirac–Pauli representation can nevertheless be advantageous for certain calculations. In the Dirac–Pauli representation,

$$\beta^{(D)} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix},$$

so the upper and lower two-component parts of a Dirac spinor are separated naturally. This makes the nonrelativistic limit particularly convenient, since the upper components of a positive-energy solution are the dominant components.

In contrast, the Weyl representation has

$$\alpha_j^{(W)} = \begin{pmatrix} \sigma_j & 0 \\ 0 & -\sigma_j \end{pmatrix},$$

which makes the two chiral components explicit. It is consequently particularly convenient for calculations involving chirality and massless fermions.

Therefore, no representation is physically preferred over the others. The different representations are related by changes of basis and give the same physical predictions. Their advantages are instead computational: the Dirac–Pauli representation is convenient for the nonrelativistic limit, while the Weyl representation is convenient for chirality and massless particles.

References

- [1] M. Flores. Lecture Slides *Physics 180: Nuclear and Particle Physics*. Digital lecture slides, 2026. Available from course website, [UVLé](#).
- [2] Hermann Weyl. Elektron und gravitation. i. *Zeitschrift für Physik*, 56:330–352, 1929. [doi:10.1007/BF01339504](#).

Appendices

Code

No code was used for this problem set

Thank you Mr. Kong

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